



CANDIDATE
NAME

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CIVICS
GROUP

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REGISTRATION
NUMBER

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H1 Biology

Paper 2 Structured Questions

8876/02

14 September 2018

2 hours

Candidates are to **answer all questions** in this question booklet.

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and registration number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use an HB pencil for any diagrams or graphs.

Do not use paper clips, highlighters, glue or correction fluid.

The use of an approved scientific calculator is expected, where appropriate.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
Section B	
7	
8	
Total	60

This document consists of **17** printed pages and **1** blank page.

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Section A

Answer **all** the questions in this section.

- 1 The synthesis of collagen is shown in Fig. 1.1.

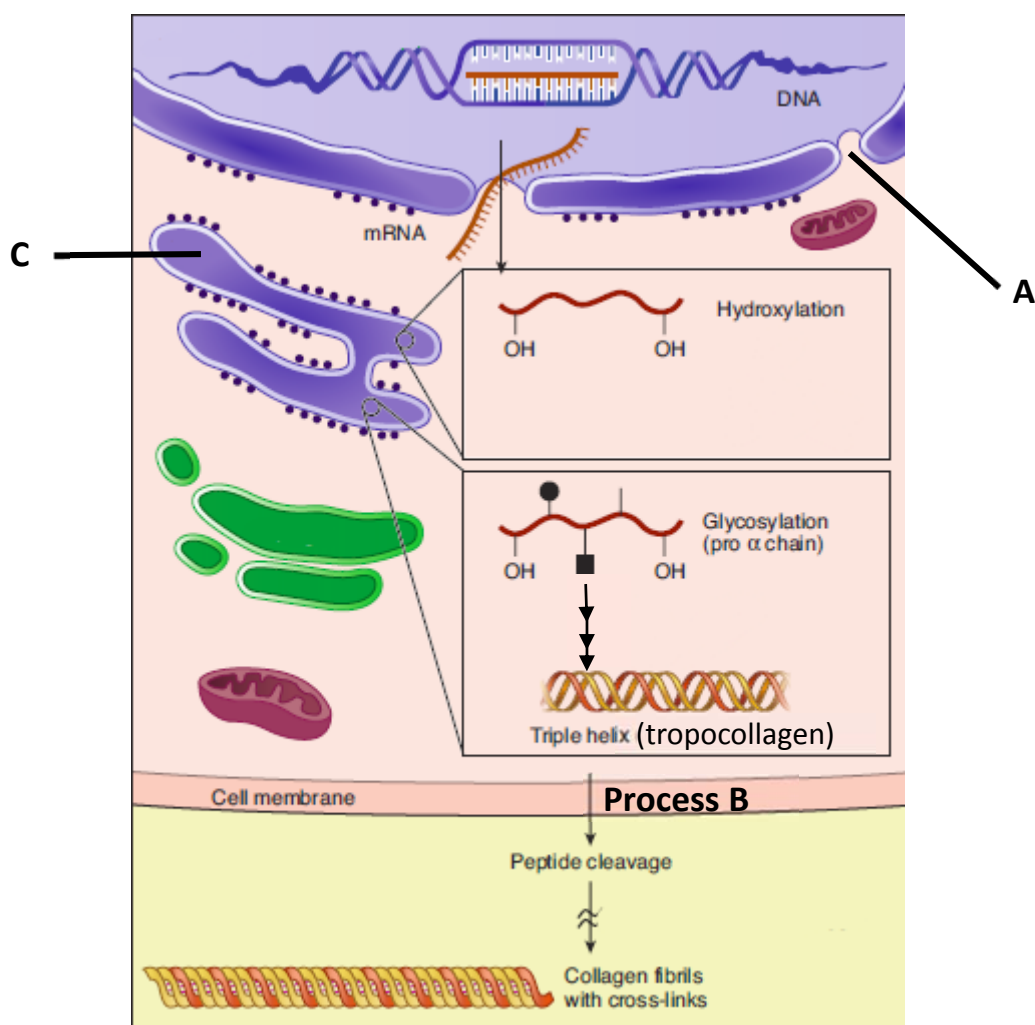


Fig. 1.1

- (a) (i) Describe two important functions of structure **A** in the synthesis of collagen.

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.....

.....

.....[2]

Structure **A** is the nuclear pore;

1. Allows mature mRNA to leave nucleus and enter cytosol for translation into collagen by ribosome;
2. Allows RNA polymerase/ ribonucleotides to enter nucleus to synthesise mRNA/tRNA during transcription of collagen gene;
3. Allows exit of mRNA to ribosomes to be translated;

4. Allows tRNA to leave nucleus for amino acid activation/ amino acid attachment for use in translation during collagen synthesis;
(any 2)

- (ii) Tropocollagen leaves the cell to be assembled to form collagen fibrils via **Process B**.
Outline **Process B**.

.....
.....
.....
.....
.....
.....[3]

1. **Process B is exocytosis;**
2. **Secretory vesicle containing tropocollagen is transported to cell surface membrane and fuses with the cell surface membrane to release tropocollagen extracellularly;**
3. **This process requires ATP;**

Fig. 1.2 shows two electron micrographs. One of the electron micrograph shows part of a normal cell while the other electron micrograph shows part of a cancer cell. The white arrows point to an organelle within the cell. The appearance of this organelle in both cell types were visibly different. The cancer cell had a higher activity than the normal cell.

Organelle (indicated by white arrows) in a normal cell

Organelle (indicated by white arrows) in a cancer cell

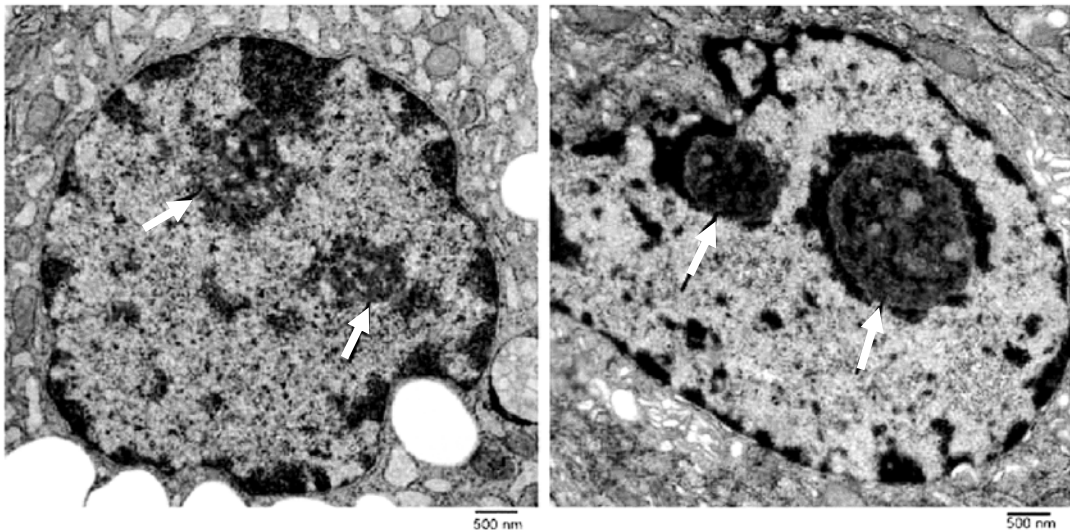


Fig. 1.2

- (b) (i) Describe the visible difference between the organelle indicated by the white arrows in Fig. 1.2, between the cancer cell and the normal cell.

.....
.....[1]

1. Nucleoli in the cancer cell were denser/darker/more darkly stained and larger;

(ii) Account for the higher activity of the cancer cell.

.....

[2]

1. Nucleoli are the sites for rRNA synthesis which are key components of ribosomes required for translation;
2. Larger and denser nucleoli in cancer cells due to increased rRNA enable cancer cell to achieve higher rates of protein synthesis to enable excessive proliferation/growth;

[Total: 8]

2 Bone marrow contains stem cells that divide by mitosis to form blood cells. The fate of a stem cell was tracked and it was recorded that during the observed duration the stem cell divided asymmetrically each time.

Fig. 2.1 shows changes in the mass of DNA in a human stem cell from bone marrow during three cell cycles.

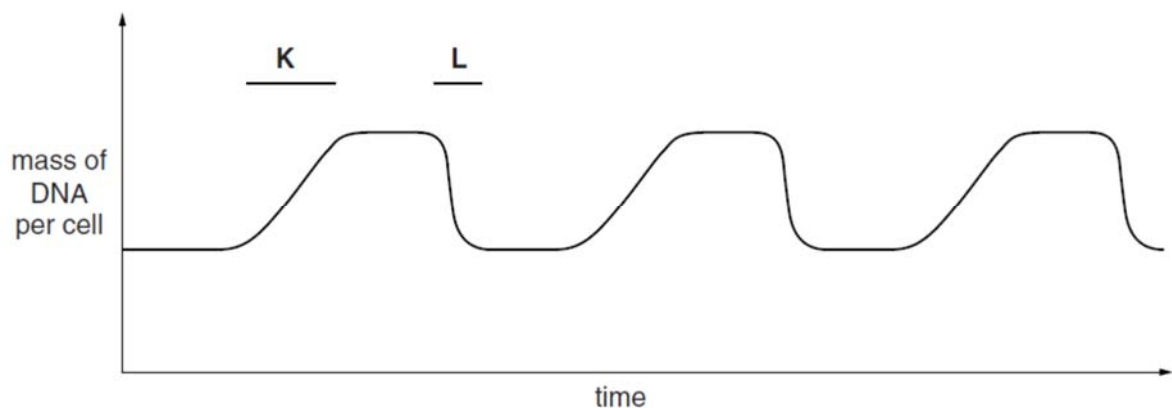


Fig. 2.1

(a) With reference to the information provided above,

- (i) Describe what happens to bring about the changes in the mass of DNA per cell at time period **K** and at time period **L**.

K

.....

L

-[2]
5. K: DNA replication/ synthesis during S phase of interphase;
6. L : cytoplasmic division during cytokinesis;

(ii) State one function of these stem cells undergoing the above type of cell division.

.....

.....[1]

1. To replace blood cells that die due to injury/ wear and tear/ disease;

(iii) The process of meiosis is significant to natural selection in evolution.

Explain this significance.

.....

.....

.....

.....[2]

1. Meiosis gives rise to genetically different haploid gametes;
2. During Prophase I of meiosis, crossing over between chromatids of homologous chromosomes would give rise to different combinations of alleles

OR

During Metaphase I and Anaphase I, independent assortment and separation of homologous chromosomes would give rise to different combinations of maternal and paternal chromosomes;

3. Resulting in genetic variation in diploid organisms leading to different phenotypes/ variation in phenotypes;
4. those with advantageous phenotypes are selected for, resulting in change in allele frequency in the population over time;

(1 and 2 + 3 / 4 OR 1+ 3 and 4)

The use of embryonic stem cells (ESCs) for stem cell therapy and research is controversial and considered by many people as unethical. Scientists have circumvented this issue through the use of induced pluripotent stem cells (iPSCs) as an alternative to ESCs.

Fig. 2.2 summarises the procedure for obtaining iPSCs and its use.

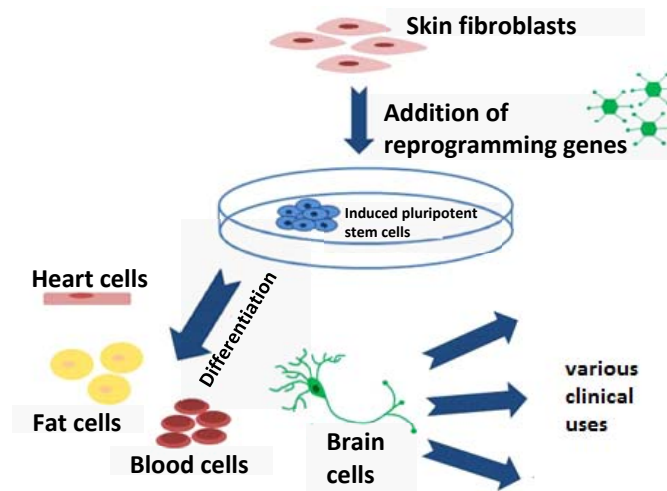


Fig. 2.2

(b) Explain the use of iPSCs is preferred over ESCs.

.....

.....

.....

-[2]
1. Since iPSCs can be derived directly from adult tissues/specialised somatic cell/ skin cells, it does not destroy any human embryos unlike ESCs;
 2. iPSCs from adult/specialised somatic cell/ skin cell can be easily obtained without risk to the donor, whereas obtaining the embryo to isolate ESCs is more invasive; (R: easy vs difficult because no elaboration)
 3. In contrast to ES cells extracted from human embryos, iPSCs derived from a patient's own cells would open the possibility of generating patient-specific cells, which will not be rejected by the immune system upon transplantation; (A: idea of iPSCs obtained from self vs embryos not from self thus contain antigens that will be rejected by immune system);
 4. Sources of obtaining iPSCs is more easily accessible compared to the source of obtaining ESCs/ obtaining the embryo; (any one, answer must make comparison with ESCs)

[Total: 7]

3 Fig. 3.1 shows part of a DNA molecule.

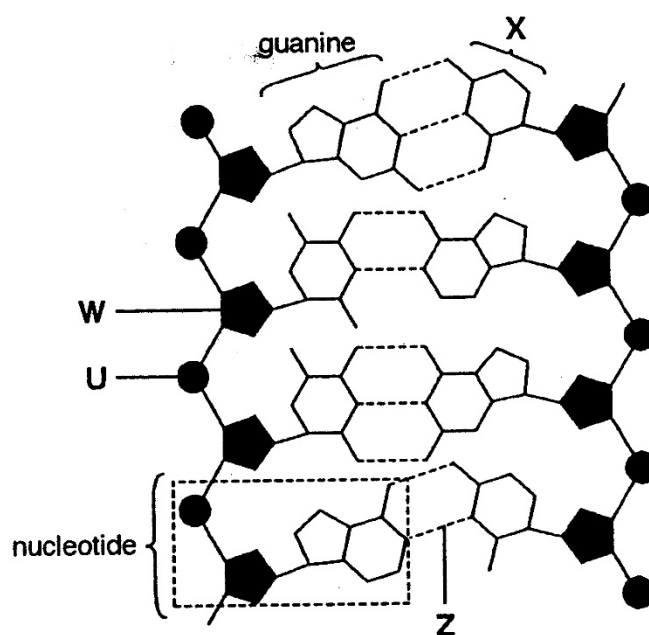


Fig. 3.1

(a) (i) Name **W** to **Z**. [2]

W

U

X

Z

1. **W = Deoxyribose (R: sugar, ribose, pentose)**

2. **U = Phosphate group (R: phosphate head)**

3. **X = Cytosine (R: nitrogenous base, pyrimidine)**

4. **Z = Hydrogen bond**

4 correct = 2 marks, 2-3 correct = 1 mark, less than 2 correct = 0 marks

(ii) Give **one** advantage of DNA having two strands.

.....

.....[1]

1. Collective/numerous hydrogen bonds between the complementary base pairs stabilise the double helical structure/molecule;

2. One strand act as a template for repair during DNA replication;

3. Protects / shields the hydrophobic nitrogenous bases from the hydrophilic medium;

- (b) Fig. 3.2 shows a linear chromosome undergoing the first round of DNA replication.

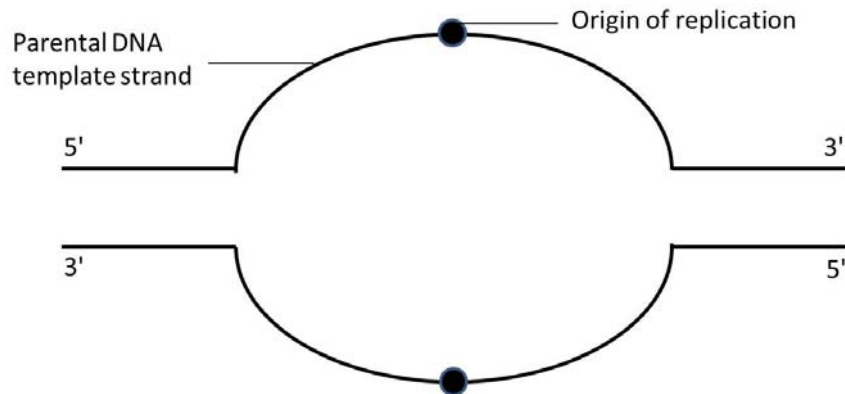
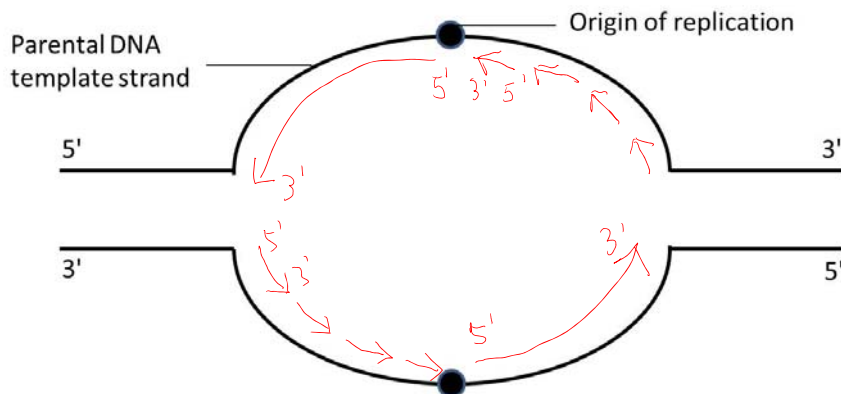


Fig. 3.2

- (i) On Fig. 3.2, draw the direction of DNA synthesis for the leading strands using solid arrows ($\rightarrow$) and for the lagging strands using dashed arrows ($- - - \rightarrow$).

[1]



1. All directions of leading and lagging strands (Okazaki fragments) correctly indicated;
2. Lagging strands must show several Okazaki fragments in correct directions;

Points 1 and 2 must be present for full credit. Not marking for 5'/ 3' labels, but arrow head needed.

- (ii) State **two** ways DNA replication is different from transcription.

.....

.....

.....

.....[2]

DNA Replication	Transcription
<u>DNA polymerase</u> used to polymerise nucleotides/synthesize new strands;	<u>RNA polymerase</u> used to polymerise nucleotides/synthesize new strands
<u>Helicase</u> used to unzip DNA strands;	<u>RNA polymerase</u> unzip DNA strands
<u>Both</u> parental strands used as <u>templates</u> ;	Only <u>one</u> strand used as <u>template</u>
<u>DNA</u> strands produced;	<u>mRNA/tRNA/rRNA</u> strand produced
<u>Deoxyribonucleotide</u> monomers;	<u>Ribonucleotide</u> monomers
Daughter DNA molecule <u>double-stranded</u> ;	mRNA molecule <u>single stranded</u>
<u>AVP</u> ;	<u>AVP</u>

- (c) Fig. 3.3 is an electronmicrograph showing the transcription of genes for ribosomal RNAs at adjacent positions along the chromosome in a eukaryotic cell.

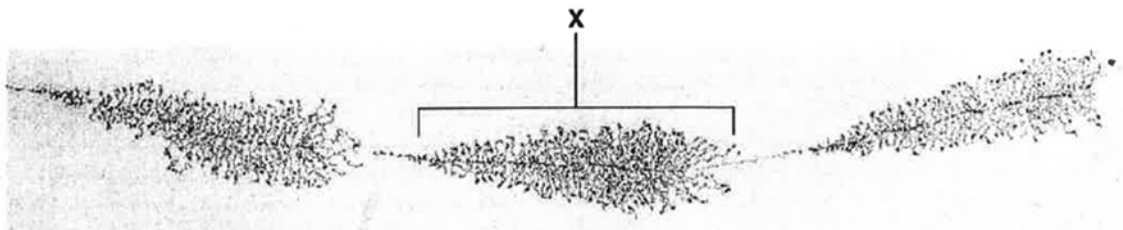


Fig. 3.3

Describe and explain the pattern of transcription visible on the part of the DNA coding for ribosomal RNA, labelled X in Fig. 3.3.

.....

.....

.....

-[2]
1. The pattern shows shorter strands of rRNA emerging on the left and with longer strands of rRNA seen on the right;
 2. This is because RNA polymerase transcribes DNA from 3' to 5' direction/ synthesises rRNA from 5' to 3' direction, hence the rRNA strands on the right (near 5' end of DNA template) were the longest (transcription started earlier);
(ora)

[Total:7]

- 4 Fig. 4.1 shows the pedigree of a family. Several members have the condition called albinism where there is a lack of normal pigment in the body.

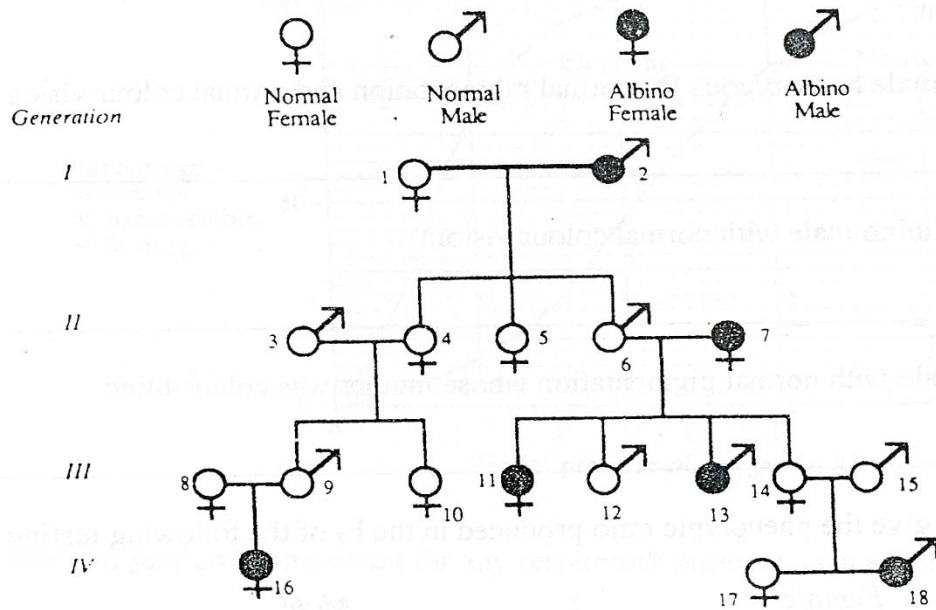


Fig. 4.1

- (a) Skin colour is determined by an autosomal gene. Individuals who are not albino possess allele A which codes for an enzyme that synthesizes pigment. Allele a, on the other hand, codes for a non-functional enzyme.

What are all the possible genotypes of the following members of the family tree? [1]

individual 3 in generation II: _____ (Aa, AA)

individual 14 in generation III: _____ (Aa)

- (b) The same family was also tested for colour blindness, a recessive sex-linked condition.

Individual 12 from generation III married a female with the same genotype as him at the A/a gene locus. Both of them are not colour blind, but one of their child is colour blind. Use a genetic diagram to show the offspring phenotypic ratio of the mating between two such individuals.

Let X^B represent the X chromosome with the dominant B allele for normal colour vision.

Let X^b represent the X chromosome with the dominant B allele for red-green colour blindness.

Let X^B represent the X chromosome with the dominant B allele for normal colour vision.

Let X^b represent the X chromosome with the dominant B allele for red-green colour blindness.

Parental phenotype	Normal pigmentation, normal colour vision female	x	Normal pigmentation, normal colour vision male	1m
Parental genotype	$AaX^B X^b$		$AaX^B Y$	
Gametes	AX^B AX^b aX^B aX^b	x	AX^B AY aX^B aY	1m

Fertilisation	Gametes	AX^B	AX^b	aX^B	aX^b	1m
		AAX^BX^B	AAX^BX^b	AaX^BX^B	AaX^BX^b	
	AX^B	normal pigmentation, normal colour vision female	normal pigmentation, normal colour vision female	normal pigmentation, normal colour vision female	normal pigmentation, normal colour vision female	
		AaX^BX^B	AaX^BX^b	aaX^BX^B	aaX^BX^b	
	aX^B	normal pigmentation, normal colour vision female	normal pigmentation, normal colour vision female	albino, normal colour vision female	albino, normal colour vision female	
		AAX^BY	AAX^bY	AaX^BY	AaX^bY	
	AY	normal pigmentation, normal colour vision male	normal pigmentation, colour blind male	normal pigmentation, normal colour vision male	normal pigmentation, colour blind male	
		AaX^BY	AaX^bY	aaX^BY	aaX^bY	
	aY	normal pigmentation, normal colour vision male	normal pigmentation, colour blind male	albino, normal colour vision male	albino, colour blind male	
Offspring genotypic ratio	$6 A_X^B_ : 2 aa X^B_ : 3 A_ X^BY : 3 A_ X^bY : 1aa X^BY : 1aa X^bY$					
Offspring phenotypic ratio	6 normal pigmentation, normal colour vision female : 2 albino, normal colour vision female : 3 normal pigmentation, normal colour vision male : 3 normal pigmentation, colour blind male : 1 albino, normal colour vision male : 1 albino, colour blind male					

1m

[Total: 5m]

- 5 A student carried out an investigation on yeast respiration using the following setup shown in Fig. 5.1. She put 5 grams of yeast into a glucose solution and put a layer of oil on top of the mixture. This layer of oil is impermeable to oxygen but not to carbon dioxide. The total volume of gas was recorded every 10 minutes for 1 hour and the results are shown in Table 5.1.

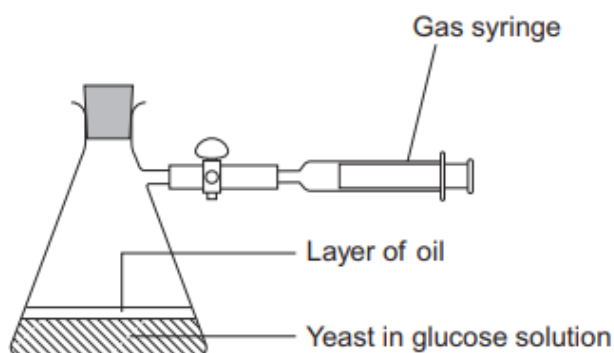


Fig. 5.1

Table 5.1

Time / minutes	Total volume of gas collected / cm ³
10	0.3
20	0.9
30	1.9
40	3.1
50	5.0
60	5.2

- (a) (i) Calculate the rate of gas production in cm³ g⁻¹ min⁻¹ during the first 40 minutes of this investigation. Show your working clearly. [1]

Answer = cm³ g⁻¹ min⁻¹

$$3.1 \text{ cm}^3 / 5\text{g} / 40\text{min} = 0.0155 \text{ OR } 0.016$$

- (ii) Suggest why the rate of gas production decreased between 50 and 60 minutes.

.....
[1]

1. **Glucose becomes a limiting factor;**
OR
2. **increase in ethanol / toxins build up causing yeast/cells to die;**
(Note: ethanol disrupts yeast cell membrane causing cell death)

- (b) The Respiratory Quotient (RQ) is defined as the ratio of carbon dioxide produced to oxygen consumed per unit time by an organism and the formula is as follows:

$$\text{RQ} = \frac{\text{volume of CO}_2 \text{ produced}}{\text{volume of O}_2 \text{ consumed}} \text{ per unit time}$$

Given that the rate of carbon dioxide produced equals to the rate of oxygen used for aerobic respiration, suggest and explain the RQ value for the setup shown in Fig. 6.1.

.....

[2]

1. RQ value more than 1 / higher than RQ value for aerobic respiration;
2. Oxygen is not consumed during anaerobic respiration but carbon dioxide is evolved during alcohol fermentation by yeast;

- (c) To show that the presence of oil has an effect on yeast respiration, the student also created another similar setup as the one shown in Fig. 6.1. All variables were kept constant in this second setup except that no oil was used.

Explain if there is any the difference in the amount of ATP produced between both setups.

.....

[3]

1. With oil: only 2 net ATP produced per glucose molecule, whereas without oil: 34/ 36/ 38 ATP produced per glucose molecule / about 17/ 18/ 19x more ATP produced for setup without oil as compared with setup with oil;

2. With oil, oxygen not present as the final electron acceptor (mark once), only glycolysis occurs to produce ATP;
3. without oil, oxygen is present as final electron acceptor (mark once), hence Krebs cycle and oxidative phosphorylation can occur to produce more ATP as well;

In another experiment, the student investigated the effect of temperature on the rate of oxygen consumption of the lizard, *Sauromalus hispidus*. The body temperature of a lizard varies with environmental temperature.

Several baby lizards were fitted with small, airtight masks that covered their heads. Air was supplied inside the mask through one tube, and collected through another. The differences between oxygen concentrations in the air supplied for inhalation and the exhaled air enabled the researchers to measure the rate of oxygen consumption of the lizards.

The rate of oxygen consumption of each baby lizard was measured at different temperatures ranging from 15 °C to 40 °C and the average measurements were computed. The results are shown in Fig. 5.2.

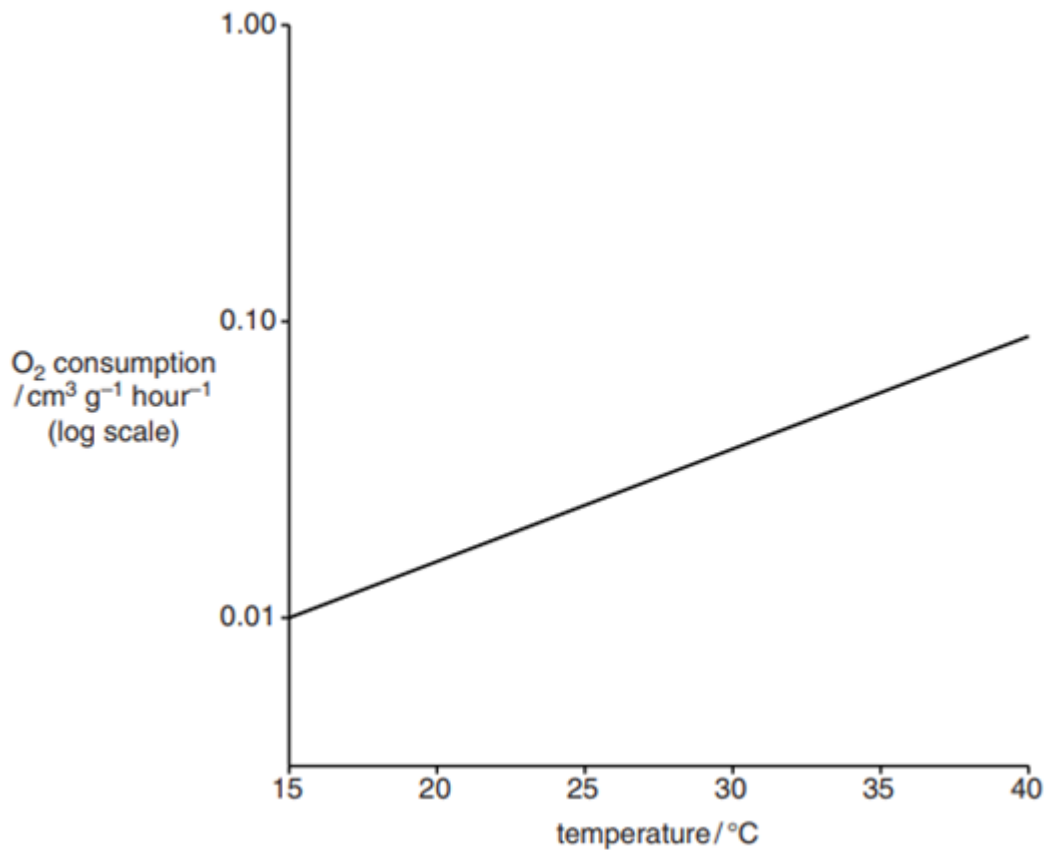


Fig. 5.2

- (d) Using Fig. 6.2, suggest and explain how temperature influences the development of baby lizards.

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.....[3]

1. As temperature increase from 15 °C to 40 °C, the rate of oxygen consumption increased from 0.01 cm⁻³ g⁻¹ hour⁻¹ to 0.09 cm⁻³ g⁻¹ hour⁻¹;
2. because increase in temperature, increases kinetic energy of respiratory enzymes and substrates, leading to increased rate of effective collisions between enzyme and substrate / increased rate of formation of enzyme-substrate complexes;
3. Hence, with an increase in respiration rate, thus providing more ATP for growth, hence rate of development of baby lizards will increase;

[Total: 10]

- 6 The theory of continental drift was proposed over 90 years ago by Alfred Wegener and is now a widely accepted theory of how the continents were formed.

The supercontinent Gondwana is shown in Fig. 6.1. The land that was to form New Zealand began to break away from the supercontinent, Gondwana around 80 million years ago and broke away totally around 65 million years ago (Fig. 6.2). Ever since, New Zealand is now composed of North Island and South Island (Fig. 6.3).

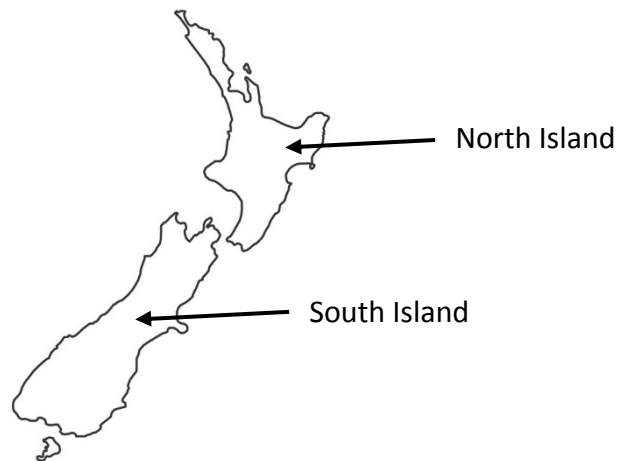
The separation of New Zealand from Gondwana has led to some species like the New Zealand kiwi bird being distantly related to the rhea (South America), ostrich (Africa) and cassowary (Australia).



Fig. 6.1



Fig. 6.2

**Fig. 6.3**

For reasons that are not apparent, New Zealand was not inhabited by many mammals. Instead, its fauna became dominated by birds and insects. Birds became the predators, the scavengers, the herbivores and the insectivores. They lived everywhere from the highest mountains to the sea. Some have lost the ability to fly and became ground dwellers.

The kakapo, kea and kaka are parrots that are currently found in New Zealand (Fig. 6.4).

The kakapo (*Stringops habroptila*) is found only in New Zealand and is flightless. Highly endangered, they were relocated to several islands in New Zealand during the 1980s and 1990s.

The kea (*Nestor notabilis*) is the world's only alpine parrot, it can be found living near forested mountain habitats - Southern Alps on South Island. Kea are not found on North Island now.

Kaka (*Nestor meridionalis*) can be found on South Island and the sub-species of Kaka (*Nestor meridionalis septentrionalis*) is found on North Island. All Kaka live in the lowlands and mid-altitude regions.



kakapo



kea



kaka

Fig. 6.4

- (a) Suggest a possible factor causing birds like the kakapo to become flightless early in their evolutionary history.

.....
[1]

1. Lack of predators / mammals that hunted the kakapo for food;
2. AVP;

Fig. 6.5 shows the phylogenetic relationships between the kakapo, kea and kaka derived from mtDNA evidence, together with the geological changes that shape the landscape of New Zealand.

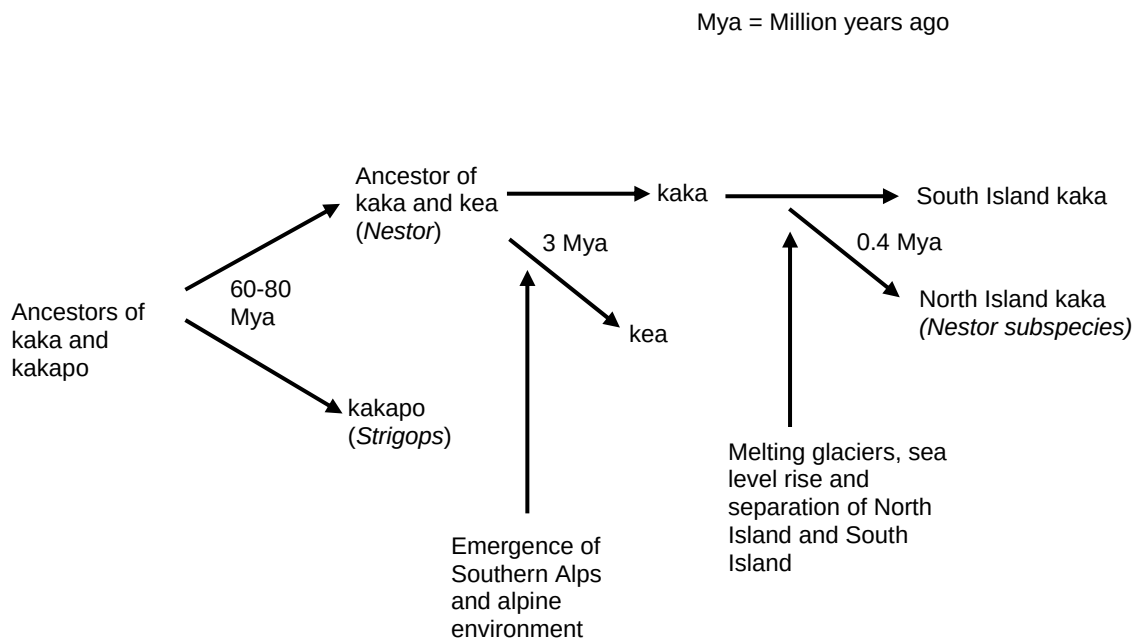


Fig. 6.5

- (b) With reference to Fig. 6.5, explain the distribution and evolution of the kea.

.....

-[5]
1. Emergence of Southern Alps / alpine environment, 3 million years ago, provided opportunity for a founder population / ancestors of the kea to exploit this habitat / niche;
 2. This sub-population was exposed to a different environment with different selection pressures compared to original population;
 3. There is existing variation in this sub-population, those birds with favourable traits / adaptations for alpine environment have a selective advantage to the local conditions and will be selected for,
 4. and will survive, reproduce and pass on their alleles to the next generation, thus increasing the frequency of favourable alleles;
 5. Those best adapted to alpine environment remain in the alps, thus gene flow is disrupted.
 6. As this sub-population evolved independently from other subpopulation, / accumulated different genetic mutations, allele frequencies changed due to genetic drift and natural selection;
 7. Over hundreds and thousands of generations, the alpine population became reproductively isolated, no longer interbreed to produce viable, fertile offspring;

- (c) With reference to Fig. 6.5, suggest a possible reason for the emergence of a sub-species of kaka on North Island.

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.....

.....

-[2]
1. North and South Island become separated due to the land being covered by sea during melting of glaciers 0.4 million years ago;
 2. Geographical isolation and independent evolution into a sub-species occurring in isolated North island kaka sub-population;

OR

3. Only a sub-species because 0.4 million years ago is insufficient time for complete (WTTE) speciation / reproductive isolation to take place;
4. Both North and South Island kaka can interbreed and produce fertile offspring if brought together;

[Total: 8]

Section B

Answer either Question 7 or 8.

Write your answers for **Q7** or **Q8** on separate pieces of the answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

7 (a) Discuss the importance of shapes fitting together in cells. [7]

1. Enzyme + substrate (max 3)

- (a) Enzyme has an active site with complementary shape to substrate;
- (b) enzymes are specific to the substrate(s) they act on; (importance)
- (c) Allows for formation of enzyme-substrate complex whereby bonds in substrate are strained (any of one e.g. of how activation energy can be lowered),
- (d) allowing for activation energy of reaction to be lowered such that biochemical reactions can occur in cells; (importance)
- (e) E.g. of enzyme catalysed reaction, e.g. aminoacyl-tRNA synthetase, rubisco, cytochrome c oxidase, peptidyl transferase (importance)

2. Cell transport (max 3)

- (a) Carrier protein has a binding site complementary in shape to molecule to be transported;
- (b) Specific carrier proteins can only bind / transport specific molecules; (importance –idea of specificity)
- (c) Binding of molecule to be transported to carrier protein induces a conformational change in carrier protein;
- (d) This allows for molecule to be transported across cell membrane; (importance)

(OR pumps + ATP)

3. Protein structure (max 3)

- (a) Haemoglobin is composed of four subunits;
- (b) Each subunit is complementary in shape to a haem group and binds to a haem group;
- (c) When one subunit binds to an oxygen, conformational change results and other subunits are able to bind oxygen with higher affinity – cooperative binding;
- (d) Allows for efficient loading of haemoglobin with oxygen (cooperative binding) / transport of oxygen (presence of haem group); (importance)

OR

Polymerisation of tubulin / ribosomal subunits (or E, P, A sites)

4. **QWC** – at least **two different examples** of shapes fitting together in cells

(b) Using named examples, describe the function of cycles in Biology. [8]

1. Krebs cycle;
 - (a) Acetyl-coA combines with OAA to produce citrate/6C compound, followed by regeneration of OAA;
 - (b) To result in the continual production of NADH and FADH₂ via a series of redox reactions; (fn)
 - (c) as well as continual synthesis of ATP via substrate-level phosphorylation; (fn)
2. Calvin cycle / light-independent reaction;
 - (a) Carbon dioxide combines with RuBP / carbon fixation, followed by the reduction of glycerate-3-phosphate to glyceraldehyde-3-phosphate and then regeneration of RuBP;
 - (b) To allow for continual formation of glucose / formation of large amounts of glucose;
 - (c) which is used for synthesis of ATP via respiration / for building of cellular components such as cellulose cell wall;
3. Cyclic photophosphorylation
 - (a) Photo-excited electrons from PS I are transferred to the ETC before returning to PS I;
 - (b) Allows for the synthesis of ATP via chemiosmosis for Calvin cycle to continue;
4. Electron transport
 - (a) Cyclical reduction of NAD, FAD / electron carriers of ETC, and oxidation of NADH, FADH₂ and reduced electron carriers of ETC; (must have)
 - (b) To allow for continual oxidation of respiratory substrate glucose, for the synthesis of ATP; (fn)
OR
 - (c) Cyclical reduction of NADP⁺ and oxidation of NADPH; (must have)
 - (d) To allow for light-dependent reaction / Calvin cycle to continue to produce glucose; (fn)
5. Mitotic cell cycle;
 - (a) Involves interphase followed by mitosis then cytokinesis;
 - (b) DNA is replicated during S phase of interphase, doubling the amount of DNA for cell division;
 - (c) Sister chromatids separate during anaphase of mitosis, DNA amount is halved after cytokinesis; (Note: mark only once for "mitosis" in point a or c)
 - (d) Results in production of genetically identical daughter cells, for growth / asexual reproduction;
 OR
6. Meiotic cell cycle
 - (a) Involves interphase followed by 2 rounds of meiotic / reduction division;
 - (b) At the end of meiosis I, haploid condition is attained;
 - (c) During fertilisation, fusion of haploid gametes restores the diploid condition;
 - (d) Allowing for chromosome number to be maintained from generation to generation;
7. QWC – at least **two to three** different cycles

8 (a) Using examples, discuss the importance of movement in cellular processes. [7]

1. Movement of chromosomes during mitosis / meiosis
 - (a) During metaphase of mitosis, chromosomes align singly along the metaphase plate;
 - (b) Following which, during anaphase of mitosis, centromeres divide and sister chromatids separate, each becoming a chromosome and are pulled to opposite poles;
 - (c) The proper separation allows the formation of genetically identical cells at the end of mitosis; (importance)
2. Transport processes (simple diffusion / facilitated diffusion / active transport)
 - (a) When electrons move down the ETC, energy released is used to pump protons from mitochondrial matrix into intermembrane space / across inner mitochondrial membrane;
OR
When electrons move down the ETC, energy released is used to pump protons from stroma into thylakoid space / across thylakoid membrane;
 - (b) During chemiosmosis, protons move from intermembrane space to mitochondrial matrix / across inner mitochondrial membrane through ATP synthase;
OR
During chemiosmosis, protons move from thylakoid space to stroma through ATP synthase / across thylakoid membrane;
 - (c) allowing ATP synthase to synthesise ATP; (importance)
3. Electron transport during respiration / photosynthesis
 - (a) Electrons move down the electron transport chain during oxidative phosphorylation;
 - (b) This allows a large amount of ATP to be synthesised per glucose molecule; (importance)
 - (c) Electrons also move down the electron transport chain during photophosphorylation;
 - (d) This allows for ATP and NADPH to be synthesised for Calvin cycle to proceed; (importance)
4. DNA replication / transcription / translation (reading of template, directionality)
 - (a) Enzyme has an
 - (b) ;
5. QWC – at least **2 to 3** examples;

NOTE: max 3m if no importance highlighted in answer throughout;

(b) Polymers have different structures and functions. Describe how such structures are related to functions. [8]

1. Storage molecules (ctrl of blood glucose, **starch**)
 - (a) **Starch** is made up of **many α glucose** joined together by **α (1, 4) glycosidic bonds**;
 - (b) Starch is **large** and therefore **insoluble in water**, and **does not affect the water potential** of cells;
 - (c) Amylose is **helical** which makes starch a **compact** molecule
OR
Amylopectin is **helical** and **branched**, making it a **compact** molecule;
 - (d) Amylopectin has **many branch points** where **amylase** can act on to **release glucose quickly**;
 - (e) Starch is therefore a good **storage** molecule;
2. Structural molecules (**cellulose**)
 - (a) **Cellulose** made up of **many β glucose** joined together by **β (1, 4) glycosidic bonds**;
 - (b) Each **β glucose** is **inverted 180° with respect to one another**, giving rise to **linear/straight** cellulose chain;
 - (c) **Intermolecular H-bonds** form between –OH groups that project out of each cellulose chain, forming **fibrils**;
 - (d) **Fibrils** associate to form **fibres**, giving rise to **high tensile strength** of cellulose;
 - (e) **Intermolecular H-bonds** form between **–OH groups** that project out of each cellulose chain, thus **unable to form H-bonds with water**, making it **insoluble in water**;
 - (f) The **large** size of cellulose also makes it **insoluble in water**;
 - (g) Cellulose is therefore a good **structural** molecule;
3. Informational molecules (**enzymes, RNA, DNA**)
 - (a) **Enzymes** are made up of **a specific number and sequence of amino acids** joined together by **peptide bonds**;
 - (b) Amino acids with **hydrophobic R groups** are buried in the **interior** while amino acids with **hydrophilic R groups** are exposed on the **exterior** of enzymes;
 - (c) giving rise to a **globular** enzyme which is **soluble in water**;
 - (d) **H-bonds, hydrophobic interactions, ionic bonds** and disulfide bonds form between **R groups** of amino acids,
 - (e) giving rise to a **specific 3D conformation** of **active site** which can only **bind** to **specific substrates** / catalyse **specific reactions** in the cell;
 - (f) **DNA** strand is made up of a specific sequence of **nucleotides** joined together by **phosphodiester bond**;
 - (g) DNA contains **genes** which code for specific **proteins**;
 - (h) Presence of **H-bonds** between **complementary base pairs** in a **double stranded** DNA molecule gives rise to a **stable** molecule;
 - (i) **Doubled stranded** nature of DNA allows one strand to serve as a **backup strand** for **DNA repair**;
 - (j) Both strands of DNA serve as **templates** for the synthesis of **complementary** DNA strand, giving rise to **identical daughter DNA molecules** via semi-conservative DNA replication;
 - (k) This makes DNA a good **storage** molecule of **genetic information**;

4. **QWC** – at least two named examples.